



UNIVERSIDAD DE ANTIOQUIA
1803

Effective two-component and two-mediator dark matter

Diego Restrepo

Instituto de Física
Universidad de Antioquia
Phenomenology Group
<http://gfif.udea.edu.co>

In collaboration with

Kimy Agudelo, Andrés Rivera and David Suárez (UdeA)



Baryon and Lepton Nonconserving Processes

Steven Weinberg (Harvard U.) (1979)

Published in: *Phys.Rev.Lett.* 43 (1979) 1566-1570

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#1 Systematic Study of One-Loop Dirac Neutrino Masses and Viable Dark Matter Candidates

Chang-Yuan Yao (Hefei, CUST), Gui-Jun Ding (Hefei, CUST) (Jul 31, 2017)

Published in: *Phys.Rev.D* 96 (2017) 9, 095004, *Phys.Rev.D* 98 (2018) 3, 039901 (erratum) • e-Print: [1707.09786](#) [hep-ph]

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Name	$SU(2)_L$	$U(1)_Y$
L	2	$-1/2$
H	2	$1/2$

$$\mathcal{L}_M = L \cdot H L \cdot H + \text{h.c}$$

Dark matter (radiative) realization $\rightarrow Z_2$

$$\mathcal{L}_D = (\nu_R)^\dagger L \cdot H + \text{h.c}$$

$$\overline{M_{N^c R^c R}} \rightarrow U(1)$$

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Scotogenic one-loop realizations

Majorana

Dirac

Systematic study of the d=5 Weinberg operator at one-loop order

#1

Florian Bonnet (Wurzburg U.), Martin Hirsch (Valencia U., IFIC), Toshihiko Ota (Munich, Max Planck Inst.), Walter Winter (Wurzburg U.) (Apr, 2012)

Published in: *JHEP* 07 (2012) 153 • e-Print: [1204.5862](#) [hep-ph]

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Models with radiative neutrino masses and viable dark matter candidates

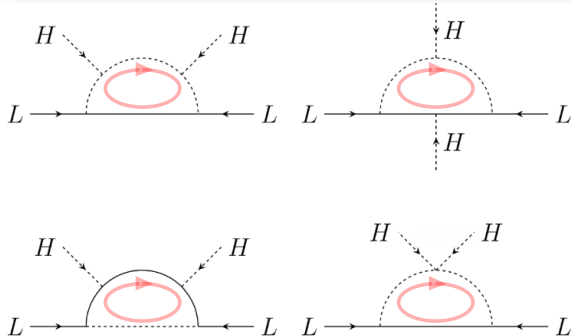
#1

Diego Restrepo (Antioquia U.), Oscar Zapata (Antioquia U.), Carlos E. Yaguna (Munster U., ITP) (Aug 16, 2013)

Published in: *JHEP* 11 (2013) 011 • e-Print: [1308.3655](#) [hep-ph]

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Pathways to Naturally Small Dirac Neutrino Masses

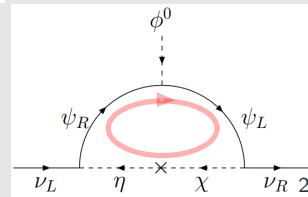
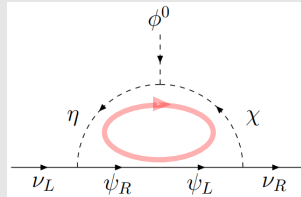
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Published in: *Phys.Lett.B* 764 (2017) 142-144 • e-Print: [1609.02538](#) [hep-ph]

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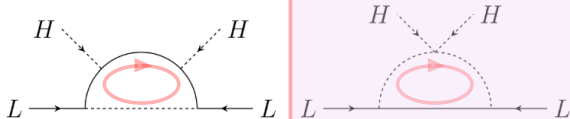
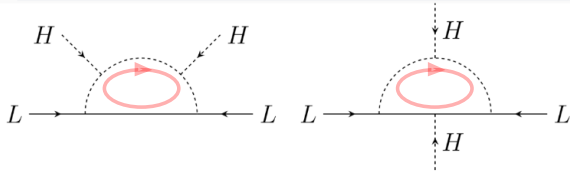
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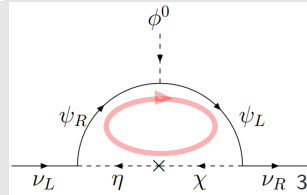
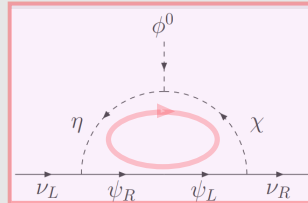
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#1
Radiative seesaw mechanism at weak scale

Zhi-jian Tao (Beijing, Inst. High Energy Phys.) (Mar, 1996)

Published in: *Phys.Rev.D* 54 (1996) 5693-5697 • e-Print: [hep-ph/9603309](https://arxiv.org/abs/hep-ph/9603309) [hep-ph]

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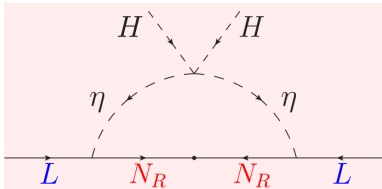
#2
Verifiable radiative seesaw mechanism of neutrino mass and dark matter

Ernest Ma (UC, Riverside) (Jan, 2006)

Published in: *Phys.Rev.D* 73 (2006) 077301 • e-Print: [hep-ph/0601225](https://arxiv.org/abs/hep-ph/0601225) [hep-ph]

pdf DOI cite claim reference search 1,539 citations

Name	$SU(2)_L$	$U(1)_Y$	Z_2
N_R	1	0	—
η	2	1/2	—

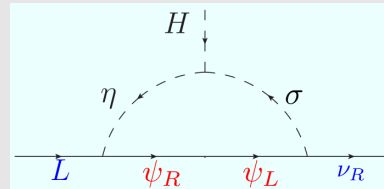
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Radiative Neutrino Mass, Dark Matter and Leptogenesis

Pei-Hong Gu (ICTP, Trieste), Utpal Sarkar (Ahmedabad, Phys. Res. Lab) (Dec, 2007)

Published in: *Phys.Rev.D* 77 (2008) 105031 • e-Print: [0712.2933](https://arxiv.org/abs/0712.2933) [hep-ph]

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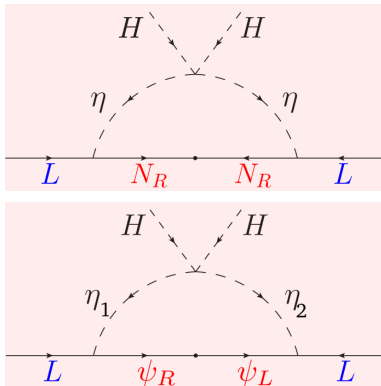
Name	$SU(2)_L$	$U(1)_Y$	Z_2	$U(1)_L$
ν_R	1	0	+	-1
ψ_R	1	0	—	-1
ψ_L	1	0	—	-1
η	2	1/2	—	0
σ	1	0	—	0



Radiative seesaw mechanism at weak scale

#1

Zhi-jian Tao (Beijing, Inst. High Energy Phys.) (Mar, 1996)

Published in: *Phys.Rev.D* 54 (1996) 5693-5697 • e-Print: [hep-ph/9603309](#) [hep-ph]
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New Scotogenic Model of Neutrino Mass with $U(1)_D$ Gauge Interaction

#1

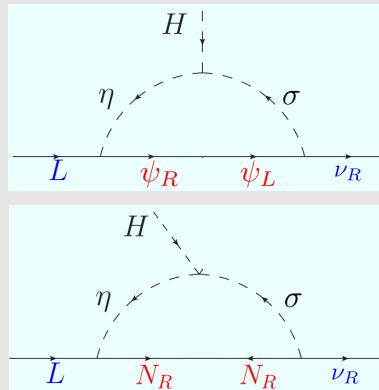
Ernest Ma (UC, Riverside), Ivica Picek (Zagreb U., Phys. Dept.), Branimir Radovčić (Zagreb U., Phys. Dept.) (Aug 24, 2013)

Published in: *Phys.Lett.B* 726 (2013) 744-746 • e-Print: [1308.5313](#) [hep-ph]
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[124 citations](#)


Dirac neutrino mass generation from a Majorana messenger

#1

Julian Calle (Antioquia U.), Diego Restrepo (Antioquia U. and IIP, Brazil), Óscar Zapata (Antioquia U.) (Sep 20, 2019)

Published in: *Phys.Rev.D* 101 (2020) 3, 035004 • e-Print: [1909.09574](#) [hep-ph]
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[DOI](#)
[cite](#)
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[reference search](#)
[15 citations](#)

Dark matter stability and Dirac neutrinos using only Standard Model symmetries

Cesar Bonilla (Munich, Tech. U.), Salvador Centelles-Chuliá (Valencia U., IFIC), Ricardo Cepedello (Valencia U., IFIC), Eduardo Peinado (Mexico U.), Rahul Srivastava (Valencia U., IFIC) (Dec 4, 2018)

Published in: *Phys.Rev.D* 101 (2020) 3, 033011 • e-Print: 1812.01599 [hep-ph]

Minimal radiative Dirac neutrino mass models

Julian Calle (Antioquia U.), Diego Restrepo (Antioquia U.), Carlos E. Yaguna (UPTC, Tunja), Óscar Zapata (Antioquia U.) (Dec 13, 2018)

Published in: *Phys.Rev.D* 99 (2019) 7, 075008 • e-Print: 1812.05523 [hep-ph]

pdf DOI cite claim reference search 46 citations

$$\mathcal{L}_5 = \frac{h}{\Lambda} (\nu_R)^\dagger L \cdot HS^* + \text{H.c.},$$

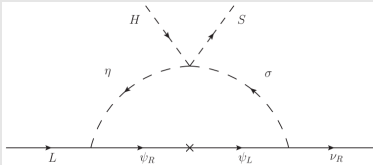


FIG. 6. T3-1-A-I ($\alpha = 0$): $B - L$ flux in the Dirac radiative seesaw.

T3-1-A-I	$(\nu_{Ri})^\dagger$	$(\nu_{Rj})^\dagger$	$(\nu_{Rk})^\dagger$	$(\psi_R)^\dagger$	ψ_L	η_a	σ_a	S
	L	$+4$	$+4$	-5	r	$-r$	$1-r$	$4-r$

Anomaly cancellation of a U(1) gauge symmetry with chiral fermions

In a fundamental theory the elementary fermions are expected to be chiral, i.e., their charges should not allow a mass term larger than the scale of spontaneous symmetry breaking. For any set of charges associated to a U(1) gauge symmetry

$$\mathbf{Z} = [Z_1, Z_2, \dots, Z_N] ,$$

The triangle anomaly with three U(1) gauge bosons on the external lines, i.e., the $[U(1)]^3$ anomaly, and with one U(1) gauge boson and two gravitons on the external lines should also be cancelled

$$\sum_{\alpha=1}^N Z_{\alpha} = 0, \qquad \sum_{\alpha=1}^N Z_{\alpha}^3 = 0, \qquad (1)$$

$U(1)_Y$: no vector-like, i.e., pairs of fields with ‘equal-but-opposite’ charges

The hypercharge for one generation in the standard model can be seen as gauge Abelian symmetry $U(1)_Y$ with 15 left-handed Weyl massless fermions with integer charges

$$Y = [1, 1, 1, 1, 1, 1, -4, -4, -4, 2, 2, 2, -3, -3, 6]$$

which satisfy

$$\sum_i Y_i = 0, \quad \sum_i Y_i^3 = 0.$$

If we introduce an scalar, H , of charge 3 we can form Yukawa couplings through the Dirac pairs

$$u_\alpha = (1, -4), (1, -4), (1, -4), \quad d_\alpha = (1, 2), (1, 2), (1, 2), \quad e = (-3, 6)$$

and the *chiral fermions* acquire Dirac masses after the EWSB. They are degenerate because the additional color symmetry. One remains massless, $\nu_L = (-3)$. The remnant Z_3 symmetry guarantees the stability of the lightest quark

Secluded gauge $U(1)_D$ without vector-like fermions:

$$\mathbf{S} = [\chi_1, \chi_2, \dots, \psi_1, \psi_2, \dots, \psi_{N'}]$$

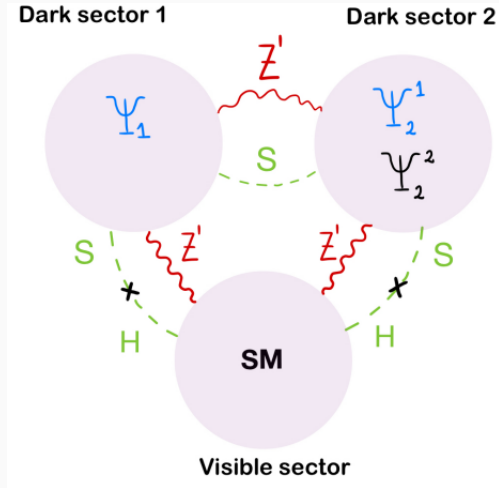
- *Dark Higgs mechanism*: Singlet dark Higgs ϕ acquires a vev and give mass to the dark photon

$$\mathcal{L} = i\psi_a^\dagger \overline{\sigma}^\mu \left(\partial_\mu - ig_D Z_\mu^D \right) \psi_a - \frac{1}{4} V_{\mu\nu} V^{\mu\nu} + \sum_{a < b} h_{ab} \psi_a \psi_b \phi^{(*)} + \text{h.c.} - V(\phi). \quad (2)$$

- S_α are the charges of SM-singlet right-handed chiral fermions with $N \geq 5$
 - χ_i *massless fermions* with $i = 1, \dots, N'$ with $N' \leq N$
 - ψ_a *multi-component dark matter*: massive after the spontaneous symmetry breaking of $U(1)_D$ with $a = N' + 1, \dots, N$
- *Larger parameter space*: Dark photon, Z_D , exclusions instead of Z'
- Z_D and ϕ masses are related by (arXiv:1610.03063)

$$h/g_D = \sqrt{2} m_\psi / m_{Z_D}.$$

Multi-component and two-mediator DM with kinetic mixing



Multi-component and two-mediator DM with kinetic mixing

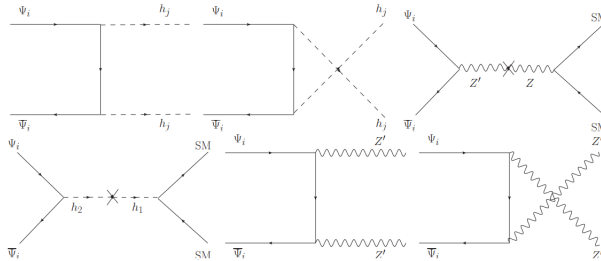


Figure 3: DM annihilation channels.

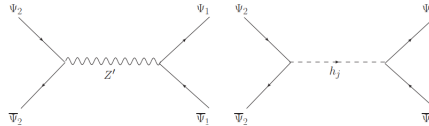


Figure 4: DM conversion channels $\Psi_1 \leftrightarrow \Psi_2$.

Decrease the number of charges to be assigned to dark matter particles, ψ_i below

$$[\chi_1, \chi_2, \dots, \psi_1, \psi_2, \dots, \psi_N]$$

Secluded case:

$$[\nu, \nu, (\nu), \psi_1, \psi_2, \dots, \psi_N]$$

$$\chi_1 \rightarrow \nu_{R1}, \dots, \chi_{N_\nu} \rightarrow \nu_{RN_\nu}, \quad 2 \leq N_\nu \leq 3,$$

$$\mathcal{L}_{\text{eff}} = h_{\nu}^{ij} (\nu_{Ri})^{\dagger} \epsilon_{ab} L_j^a H^b \left(\frac{\phi^*}{\Lambda} \right)^{\delta} + \text{H.c.}, \quad \text{with } i, j = 1, 2, 3,$$

ϕ is the complex singlet scalar responsible for the SSB of the anomaly-free gauge symmetry and give mass to all ψ_a

$$\phi = -\frac{\nu}{\delta},$$

Decrease the number of charges to be assigned to dark matter particles, ψ_i below

$$[\chi_1, \chi_2, \dots, \psi_1, \psi_2, \dots, \psi_N]$$

Secluded case:

$$[5, 5, -3, -2, 1, -6]$$

$$\chi_1 \rightarrow \nu_{R1}, \chi_2 \rightarrow \nu_{R2}, \quad N_\nu = 2,$$

$$\mathcal{L}_{\text{eff}} = h_\nu^{aj} (\nu_{Ra})^\dagger \epsilon_{bc} L_j^b H^c \left(\frac{\phi^*}{\Lambda} \right) + \text{H.c.}, \quad \text{with } j = 1, 2, 3,$$

ϕ is the complex singlet scalar responsible for the SSB of the anomaly-free gauge symmetry and give mass to all ψ_a

$$\phi = -\nu = -5,$$

Minimal secluded model with D-5 effective Dirac neutrino masses

$$\mathcal{L} = i\psi_i^\dagger \not{D}\psi_i - \frac{1}{4}V_{\mu\nu}V^{\mu\nu} + \sum_{i<j} h_{ij}\psi_i\psi_j\phi^{(*)} + \text{h.c.} - V(\phi). \quad (3)$$

multi-component DM with two Dirac-fermion DM particles

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$$\mathbf{z} = [5, 5, -3, -2, 1, -6] \rightarrow \phi = -5 \rightarrow [(5, 5), (-3, -2), (1, -6)] \quad (4)$$

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$$\mathbf{z} = [5, 5, -3, -2, 1, -6] \rightarrow \phi = -5 \rightarrow [(5, 5), (-3, -2), (1, -6)] \quad (4)$$

$$\mathcal{L} \subset h_{(-3,-2)}\psi_{-3}\psi_{-2}\phi^* + h_{(1,-5)}\psi_1\psi_{-6}\phi^* + \text{h.c.} \quad (5)$$

Decrease the number of charges to be assigned to dark matter particles, ψ_i below

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give mass to all ψ_a . ν_R remains massless

$$\phi = -\frac{\nu}{\delta},$$

Multi-component dark matter model with two-mediators: Z_D and ϕ

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ϕ is dark Higgs responsible for the SSB of the anomaly-free gauge symmetry and give mass to all ψ_a . ν_R remains massless

$$\phi = -\nu = -5,$$

Two-component dark matter model with two-mediators: Z_D and ϕ

Minimal secluded model with D-5 effective Dirac neutrino masses

We assume decoupled inert scalar (vev-less) sector

$$\mathcal{L} = i\psi_i^\dagger \not{D} \psi_i - \frac{1}{4} V_{\mu\nu} V^{\mu\nu} + \sum_{i < j} h_{ij} \psi_i \psi_j \phi^{(*)} + \text{h.c.} - V(\phi).$$

two-component and two-mediator DM with two Dirac-fermion DM particles

Minimal secluded model with D-5 effective Dirac neutrino masses

We assume decoupled inert scalar (vev-less) sector

$$\mathcal{L} = i\bar{\psi}_i \not{D} \psi_i - \frac{1}{4} V_{\mu\nu} V^{\mu\nu} + \sum_{i < j} h_{ij} \bar{\psi}_i \psi_j \phi^{(*)} + \text{h.c.} - V(\phi).$$

two-component and two-mediator DM with two Dirac-fermion DM particles

$$z = [5, 5, -3, -2, 1, -6] \rightarrow \phi = -5 \rightarrow [(5, 5), (-3, -2), (1, -6)]$$

Minimal secluded model with D-5 effective Dirac neutrino masses

We assume decoupled inert scalar (vev-less) sector

$$\mathcal{L} = i\bar{\psi}_i \not{D} \psi_i - \frac{1}{4} V_{\mu\nu} V^{\mu\nu} + \sum_{i < j} h_{ij} \bar{\psi}_i \psi_j \phi^{(*)} + \text{h.c.} - V(\phi).$$

two-component and two-mediator DM with two Dirac-fermion DM particles

$$z = [5, 5, -3, -2, 1, -6] \rightarrow \phi = -5 \rightarrow [(5, 5), (-3, -2), (1, -6)]$$

$$\mathcal{L} \subset h_{(-3, -2)} \bar{\psi}_{-3} \psi_{-2} \phi^* + h_{(1, -6)} \bar{\psi}_1 \psi_{-6} \phi^* + \text{h.c.}$$

Minimal secluded model with D-5 effective Dirac neutrino masses

We assume *a minimal* decoupled inert scalar (vev-less) sector

$$\mathcal{L} = i\psi_i^\dagger \not{D}\psi_i - \frac{1}{4}V_{\mu\nu}V^{\mu\nu} + \sum_{i<j} h_{ij}\psi_i\psi_j\phi^{(*)} + \text{h.c.} - V(\phi).$$

two-component, two-mediator DM with two-flavor scotogenic sector

$$\mathbf{z} = [-9, -9, -9, 10, 10, -1, -1, 4, 5] \rightarrow \phi = 9 \rightarrow [(-9, -9, -9), (10, -1), (10, -1), (4, 5)]$$

$$\mathcal{L} \subset \underbrace{\sum_{a,b=1}^2 h_{(10a,-1b)}\psi_{10}\psi_{-1}\phi}_{\text{multi-flavor scotogenic DM}} + h_{(4,5)}\psi_4\psi_5\phi^* + \text{h.c.} \quad (6)$$

multi-flavor scotogenic DM

September 24, 2021

Dataset

Open Access

Set of N integers between -30 and 30 with sum and cubic sum up to zero for $4 < N < 13$

Diego Restrepo

Anomalies

Solutions obtained with the python package: [anomalies](#) based on the method to find anomaly free solutions of the standard model extended with an Abelian Dark Symmetry with N right-handed singlet chiral fields described in [arXiv:1905.13729](#) [PRL]:

Data scheme

- 'l': integer lists → input to obtain the 'solution' by using the [anomalies](#) package
- 'k': integer lists → input to obtain the 'solution' by using the [anomalies](#) package
- 'solution': list → of integers, z_i which satisfy $\sum_{i=1}^N z_i = 0$ and $\sum_{i=1}^N z_i^3 = 0$.
- 'n': integer → number of integers in 'solution', N .

USAGE

```
#Example of JSON file usage in Python with pandas (see also json module)
>>> import pandas as pd
>>> df=pd.read_json('solutions.json.gz')
>>> df[:2]
   l      k      solution gcd n
0  [1, 2]  [0, -3]  [1, 5, -7, -8, 9]  1  5
1  [-2, -1] [0, -1]  [2, 4, -7, -9, 10]  1  5
```

Data:

2 296 615 solutions with $5 \leq N \leq 12$ integers until 'j32' [JSON]

141

views

351

downloads

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Keyword(s):

[Anomaly free](#) [Diophantine equations](#) [Abelian symmetry](#)
[Gauge Symmetry](#)

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Versions

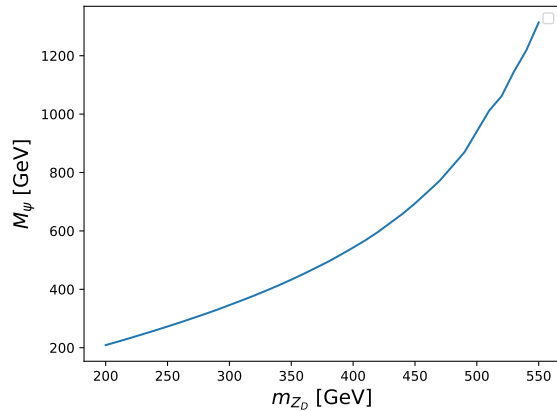
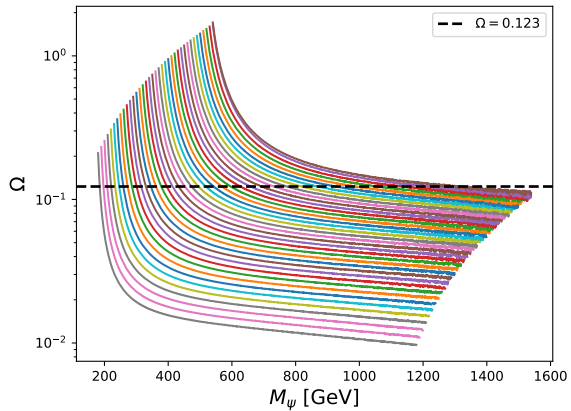
Version v2

Sep 24, 2021

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Solutions ~ 1000	N	ν	δ	ϕ	N_D	N_M	G_D	G_M
(5, 5, -2, -3, 1, -6)	6	5	1	-5	2	0	1	0
(-9, -9, -9, 10, 10, -1, -1, 4, 5)	9	9	1	-9	3	0	2	0
(1, -2, 3, 4, 6, -7, -7, -7, 9)	9	-7	1	7	3	0	1	0
(1, -2, -2, 3, 3, -4, -4, 6, 6, -7)	10	6	1	-6	3	2	2	2
(1, -2, -2, 3, 4, -5, -5, 7, 7, -8)	10	-5	1	5	4	0	2	0
(1, -2, -2, 3, 5, -6, -6, 8, 8, -9)	10	-6	1	6	4	0	2	0
(2, 2, 3, 4, 4, -5, -6, -6, -7, 9)	10	2	1	-2	4	2	2	2
(1, 1, 5, 5, 5, -6, -6, -6, -9, 10)	10	1	1	-1	4	0	3	0
(2, 2, 4, 4, -7, -7, -9, -9, 10, 10)	10	10	2	-5	3	0	2	0
(1, 2, 2, -3, 6, 6, -8, -8, -9, 11)	10	-8	1	8	4	1	2	1
(1, -2, -3, 5, 6, -8, -9, 11, 11, -12)	10	11	1	-11	4	0	1	0
(1, 1, -3, 4, 4, -7, 8, -10, -10, 12)	10	-10	2	5	4	0	2	0
(1, 1, -2, -2, -4, 6, -10, 11, 12, -13)	10	-2	1	2	3	2	1	2
(3, 4, 4, 4, 4, -5, -8, -8, -11, 13)	10	-8	1	8	2	4	1	4
(4, 4, 5, 6, 6, -9, -10, -10, -11, 15)	10	6	1	-6	4	0	2	0
(1, -2, -4, 7, 7, -10, -12, 14, 14, -15)	10	14	1	-14	3	2	1	2
(1, 2, 2, -3, 4, -6, 12, -13, -14, 15)	10	2	1	-2	4	1	1	1
(1, 4, 4, -7, 8, 8, -9, -12, -12, 15)	10	8	1	-8	4	2	2	2
(1, 2, 2, -9, -9, 16, 16, 17, -18, -18)	10	-18	1	18	3	2	2	2
(1, -3, -6, 7, -10, 11, -16, 18, 18, -20)	10	18	2	-9	4	0	1	0
(1, -4, 5, -6, -6, 10, -14, 15, 20, -21)	10	-6	1	6	4	0	1	0
(2, -3, -6, 7, 12, -14, -14, 17, 20, -21)	10	-14	1	14	4	1	1	1
(3, 6, 6, -7, 8, 8, -14, -14, -17, 21)	10	-14	1	14	4	1	2	1
(8, 8, 9, 10, 10, -13, -18, -18, -27, 31)	10	-18	1	18	4	1	2	1

$M_2 \gg M_1$ and $g_D = 0.1$



Conclusions

